

Education

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology Kanpur	8.4/10
2021	ICSE(XII)	St. Basil's School, Basti	96.40%
2019	ICSE(X)	St. Basil's School, Basti	94.67%

Scholastic Achievements

- Secured **All India Rank 2188** in **Joint Entrance Examination (JEE) Advanced** among 1.5 Lakh candidates (2021)
- Among the top **1 percent** of 12 lakh candidates appearing for **Joint Entrance Examination (JEE) Mains** (2021)
- Secured **1st rank** in Basti **district** in **ICSE** board-Science division (Physics, Chemistry, Math, Computer Application) (2020)

Key Projects

- Automatic Registration of LiDAR Point Cloud** (*SURGE, Prof Salil Goel, CE IIT Kanpur*) (May'23 - Jul'23)
 - Researched 15+ papers on various forms of **Iterative Closest Point algorithm** for **Lidar Point Cloud Registration**
 - Implemented RMS-ICP that uses techniques to improve 80% efficiency of algorithm to improve computational issues
 - Grasped multiple Data manipulation and visualization libraries like **PCL, Eigen, TensorFlow, OpenCV**, etc.
 - Developed a robust C++ program tested on 'ASL labs Laser Registration Datasets' to return corrected 3-D registration
- Infrastructure Asset Management**, *Society of Civil Engineering, IIT Kanpur* (May'23 - Jul'23)
 - Learned about **Mathematical model Development** for Cost Optimisation and Asset Management of Infrastructure
 - Explored **Continuous** and **Discrete Optimisation** and tackled **11+ problems** based on **Convex programming**
 - Analyzed Algorithms like Branch & bound, Branch & Price along with Karmakar and simplex Interior point method
 - 10+ mathematical models** for Multi-Objective Knapsack, Vehicle routing, etc were designed using **IBM Cplex**
- Problems in Combinatorics**, *Stamatics, IIT Kanpur* (May'23 - Jul'23)
 - Probed a wide range of problems on enumerative and probabilistic combinatorial mathematics and graphs theory
 - Utilized techniques like generating functions, recurrence relations, combinatorial identities, and probabilistic methods
 - Assimilated catalan number and used the counting via bijections method to solve Richard stanley's number problems
 - Presented analysis of Delanoy and Schroder paths and correspondence establishment by bijection and enumeration
- Deep Dive into Web 3.0**, *Programming Club, IIT Kanpur* (May'22 - Jul'22)
 - Studied about Web 3.0, the concept of Decentralisation, cryptocurrency, and Non-fungible Tokens, Blockchain
 - Learned about cryptography, cybersecurity and different Encryption techniques like Hash functions (SHA256) etc.
 - Deployed Smart contracts in solidity on Remix IDE for implementing a data retrieval based on Proof of Work consensus

Positions of Responsibility

- Manager, Events**, *Public Policy & Opinion Cell, IIT Kanpur* (Apr'23 - Present)
 - Spearheaded a team of **31 secretaries** and planned a budget of 5 Lacs+ for 10+ events like UPSC Talks, Niti Quiz, etc.
 - Hosted a national case study competition for **270+** on **Unstop** with **IIMC, IIML, IIFTK, BITS PILANI, IIMU**
- Project Mentor**, *Public Policy and Opinion Cell, IIT Kanpur* (Mar'23 - Apr'23)
 - Mentored a project- **Analyzing press Freedom Index** with 30+ mentees offered under public policy and opinion cell
 - Composed extensive study on data about worldwide Press Freedom using ML algorithms and various python libraries
- Secretary, Student Internship Program** *Ecell, IIT Kanpur* (Jun'22 - Apr'23)
 - Compiled list of emerging startups and unicorns in India and conducted end-to-end recruitment for Start-up Internship
 - Finalised companies of Software, Engineering, Management, Design, Finance etc, for internship opportunities for students
- Student Guide and Academic Mentor**, *Counseling Service, IIT Kanpur* (Nov'22 - Jun'23)
 - Cooperated in the successful implementation of the Orientation Programme conceptualized for **1250+ Freshers**
 - Provided Academic mentorship to 300+ freshers for course-**Fundamentals** of **Computing** with doubt clearing sessions

Skills

- Programming Languages:** C, C++, LaTeX, Python, Java, HTML, CSS, Javascript, Solidity
- Software and Libraries:** Git, Fusion 360, Remix- Ethereum lite, PCL, Eigen etc.
- Management** Leadership, Planning and Strategy, Communication, Decision Making, Negotiation

Relevant Courses

Fundamental of Computing*	Applied probability and Statistics*	Mechanics of solids*
Sustainable Built Environment*	Application of Psychology*	Introduction to Electronics*

(*) - A grade for outstanding performance

Extra-Curricular Activities

- Served an year long tenure as **Joint Secretary** in the **Student Office** at St. Basil's School, Basti (2016-17)
- Awarded prizes in solo singing, basketball, football, badminton and athletics competitions at school and district level events